Graded Assignment

(Session: Even Semester 2025**)**

Subject: Mobile Application Development (CSE3709) **Course:** B.Tech. (CSE)

Batch:2023-27 **Submission Deadline:** 19-April-2025 (10:00 PM) **Weightage:** 20 Marks

Section: _____CSE-5_____ **Name:** ____Vaibhav Jain_____ **Enrolment No:** _____230697_____

Guidelines for the assignment:

- Weightage of this graded assignment is 20% of the 100 marks (i.e. twenty marks).

- The assignment must be submitted in the .pdf file as GA_Section_UniqueID_FirstName_LastName.pdf. (*Note:* use the same file and start typing from the second page; then generate the PDF)

Example: ***GA_CSE1_230111_Vinay_Kumar.pdf***

- If the question is a programming question, the complete running code (Java and XML) should be attached to this file along with the output screenshot. The code must be in the text format, not an image, so it can be tested by running. The code should be well documented (comments) and formatted (indentation).

- Answer all the questions in the order of the questions.

- Your section, name and roll number should be written at the beginning of the assignment.

- Plagiarism will be punished, and zero marks will be awarded for cheating. **- Submission after the deadline will not be entertained.**

- You can use the real device to run the programs if required.

- This assignment contains eight questions where each question carries equal marks, i.e. 2.5 marks.

Q1. Create a basic Android application that functions as a music player, allowing users to play songs in their preferred format.

Q2. Build a simple Android application that serves as a media player, enabling playback of videos in a chosen format.

Q3. Design a basic Android application that records videos and saves them to the smartphone's storage.

Q4. Develop an Android application that showcases how to send and receive notifications.

Q5. Create a simple Android app that captures images using the smartphone's camera and stores them in the device's memory.

Q6. Build an Android application that demonstrates how to send and receive emails. Q7.

Develop a basic Android app that illustrates the process of making and receiving phone calls.

Q8. Create an Android application that demonstrates the sending and receiving of SMS messages.

Question 1

MainActivity.java

```
package com.example.musicplayer;


import android.media.MediaPlayer;

import android.os.Bundle;

import android.widget.Button;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


    private MediaPlayer mediaPlayer;

    private Button playButton, pauseButton, stopButton;


    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);
```

```
setContentView(R.layout.activity_main);


playButton = findViewById(R.id.playButton);
pauseButton = findViewById(R.id.pauseButton);
stopButton = findViewById(R.id.stopButton);


// Load music from res/raw/sample_music.mp4
mediaPlayer = MediaPlayer.create(this, R.raw.sample_music);


playButton.setOnClickListener(v -> {
    mediaPlayer.start();
    Toast.makeText(this, "Playing Music", Toast.LENGTH_SHORT).show();
});


pauseButton.setOnClickListener(v -> {
    if (mediaPlayer.isPlaying()) {
        mediaPlayer.pause();
        Toast.makeText(this, "Music Paused", Toast.LENGTH_SHORT).show();
```

```
        }
    });

    stopButton.setOnClickListener(v -> {
        if (mediaPlayer != null) {
            mediaPlayer.stop();
            mediaPlayer = MediaPlayer.create(this, R.raw.sample_music); // Re-initialize
            Toast.makeText(this, "Playback Stopped", Toast.LENGTH_SHORT).show();
        }
    });
}

@Override
protected void onDestroy() {
    super.onDestroy();
    if (mediaPlayer != null) {
        mediaPlayer.release();
        mediaPlayer = null;
```

```
        }
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp"
    android:background="@android:color/white">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Music Player"
        android:textSize="22sp"
```

```
    android:textStyle="bold"
    android:textColor="#000000"
    android:layout_marginBottom="20dp" />

<Button
    android:id="@+id/playButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Play" />

<Button
    android:id="@+id/pauseButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="12dp"
    android:text="Pause" />

<Button
```

```
        android:id="@+id/stopButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="12dp"
        android:text="Stop" />
</LinearLayout>
```

AndroidManifest.xml

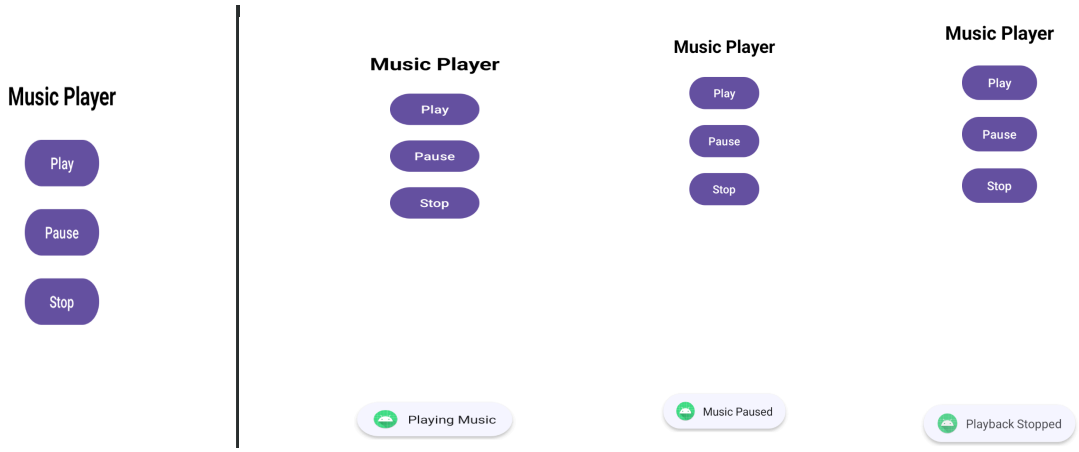
```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.musicplayer">

    <application
        android:allowBackup="true"
        android:label="MusicPlayer"
        android:icon="@mipmap/ic_launcher"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.MusicPlayer">
```

```
        <activity android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>

    </application>
</manifest>
```



Question 2

MainActivity.java

```
package com.example.videoplayer;


import android.net.Uri;

import android.os.Bundle;

import android.widget.Button;

import android.widget.VideoView;

import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


    VideoView videoView;

    Button playButton, pauseButton;


    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);
```

```
setContentView(R.layout.activity_main);


videoView = findViewById(R.id.videoView);
playButton = findViewById(R.id.playButton);
pauseButton = findViewById(R.id.pauseButton);


// Set video path from raw folder
String videoPath = "android.resource://" + getPackageName() + "/" + R.raw.sample_video;
videoView.setVideoURI(Uri.parse(videoPath));


// Play button listener
playButton.setOnClickListener(v -> videoView.start());


// Pause button listener
pauseButton.setOnClickListener(v -> {
    if (videoView.isPlaying()) {
        videoView.pause();
    }
```

```
        });
    }
}
```

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="16dp">

    <VideoView
        android:id="@+id/videoView"
        android:layout_width="match_parent"
        android:layout_height="300dp"
        android:layout_marginBottom="16dp" />

    <Button
```

```
        android:id="@+id/playButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Play" />

    <Button
        android:id="@+id/pauseButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Pause"
        android:layout_marginTop="8dp" />
</LinearLayout>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

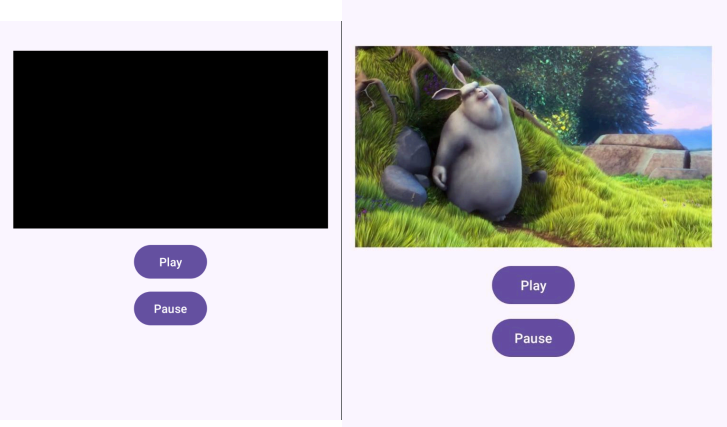
    <application
```

```
android:allowBackup="true"
android:dataExtractionRules="@xml/data_extraction_rules"
android:fullBackupContent="@xml/backup_rules"
android:icon="@mipmap/ic_launcher"
android:label="@string/app_name"
android:roundIcon="@mipmap/ic_launcher_round"
android:supportsRtl="true"
android:theme="@style/Theme.VideoPlayer"
tools:targetApi="31">
<activity
    android:name=".MainActivity"
    android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.MAIN" />


        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
</activity>
```

</application>

</manifest>



Question 3

MainActivity.java

```
package com.example.videorecorder;


import android.Manifest;

import android.content.Intent;

import android.content.pm.PackageManager;

import android.net.Uri;

import android.os.Bundle;
```

```
import android.provider.MediaStore;

import android.widget.Toast;


import androidx.annotation.Nullable;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.app.ActivityCompat;

import androidx.core.content.ContextCompat;


public class MainActivity extends AppCompatActivity {


    private static final int VIDEO_REQUEST_CODE = 200;

    private static final int PERMISSION_REQUEST_CODE = 100;

    private Uri capturedVideoUri;


    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);
```

```
        // Ask camera permission if not already given
        if (ContextCompat.checkSelfPermission(this, Manifest.permission.CAMERA)
                != PackageManager.PERMISSION_GRANTED) {
            ActivityCompat.requestPermissions(this,
                    new String[]{Manifest.permission.CAMERA},
PERMISSION_REQUEST_CODE);
        }
    }

    public void recordVideo(android.view.View view) {
        Intent videoIntent = new Intent(MediaStore.ACTION_VIDEO_CAPTURE);
        if (videoIntent.resolveActivity(getPackageManager()) != null) {
            startActivityForResult(videoIntent, VIDEO_REQUEST_CODE);
        } else {
            Toast.makeText(this, "No compatible app found to record video",
Toast.LENGTH_SHORT).show();
        }
    }
```

```
    @Override
    protected void onActivityResult(int requestCode, int resultCode, @Nullable Intent data) {
        super.onActivityResult(requestCode, resultCode, data);

        if (requestCode == VIDEO_REQUEST_CODE && resultCode == RESULT_OK && data
!= null) {
            capturedVideoUri = data.getData();
            Toast.makeText(this, "Video Recorded: " + capturedVideoUri.toString(),
Toast.LENGTH_SHORT).show();

            // Play the recorded video using default media player
            Intent playIntent = new Intent(Intent.ACTION_VIEW, capturedVideoUri);
            playIntent.setDataAndType(capturedVideoUri, "video/mp4");
            startActivity(playIntent);
        } else {
            Toast.makeText(this, "Recording cancelled or failed", Toast.LENGTH_SHORT).show();
        }
    }
```

}

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#F0F0F0"
    android:padding="24dp">

    <TextView
        android:id="@+id/titleText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Video Recorder App"
        android:textSize="24sp"
        android:textStyle="bold"
```

```
        android:textColor="#333"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="36dp" />

    <Button
        android:id="@+id/btnRecordVideo"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Record Video"
        android:onClick="recordVideo"
        app:layout_constraintTop_toBottomOf="@id/titleText"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="50dp" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.videorecorder">

    <!-- Permission to use the device camera -->
    <uses-permission android:name="android.permission.CAMERA" />
    <!-- Permission to write to external storage (optional, depends on target SDK) -->
    <uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"
        android:maxSdkVersion="28" />
    <uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE" />

    <application
        android:allowBackup="true"
        android:label="VideoRecorder"
        android:icon="@mipmap/ic_launcher"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
```

```
        android:theme="@style/Theme.AppCompat.Light.NoActionBar">

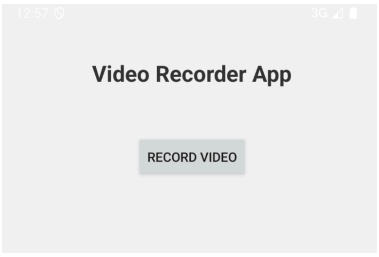

        <activity android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>

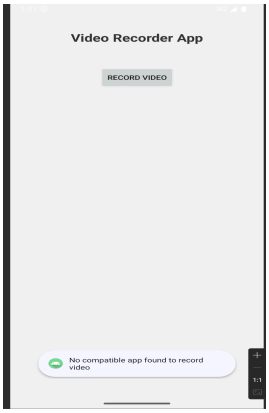

    </application>


</manifest>
```

file_paths.xml

```
<?xml version="1.0" encoding="utf-8"?>
<paths xmlns:android="http://schemas.android.com/apk/res/android">
    <external-files-path name="videos" path="Movies/" />
</paths>
```





Question 4

MainActivity.java

```
package com.example.notifications;


import android.Manifest;

import android.content.pm.PackageManager;

import android.os.Bundle;

import android.telephony.SmsManager;

import android.view.View;

import android.widget.Button;

import android.widget.EditText;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.app.ActivityCompat;


public class MainActivity extends AppCompatActivity {

    EditText phoneEditText, messageEditText;

    Button sendButton;
```

```
private static final int PERMISSION_REQUEST_CODE = 1;


@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);


    phoneEditText = findViewById(R.id.phoneEditText);
    messageEditText = findViewById(R.id.messageEditText);
    sendButton = findViewById(R.id.sendButton);


    // Request permissions for sending and receiving SMS
    ActivityCompat.requestPermissions(this, new String[]{
            Manifest.permission.SEND_SMS,
            Manifest.permission.RECEIVE_SMS,
            Manifest.permission.READ_SMS
    }, PERMISSION_REQUEST_CODE);
```

```
        sendButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String number = phoneEditText.getText().toString();
                String message = messageEditText.getText().toString();
                if (!number.isEmpty() && !message.isEmpty()) {
                    SmsManager smsManager = SmsManager.getDefault();
                    smsManager.sendTextMessage(number, null, message, null, null);
                    Toast.makeText(MainActivity.this, "SMS Sent!", Toast.LENGTH_SHORT).show();
                } else {
                    Toast.makeText(MainActivity.this, "Enter number and message",
Toast.LENGTH_SHORT).show();
                }
            }
        });
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    android:gravity="center">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="SMS Sender"
        android:textSize="24sp"
        android:gravity="center"
        android:padding="16dp"/>

    <EditText
        android:id="@+id/phoneEditText"
        android:layout_width="match_parent"
```

```
    android:layout_height="wrap_content"
    android:hint="Enter Phone Number"
    android:inputType="phone"/>

<EditText
    android:id="@+id/messageEditText"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Message"
    android:inputType="textMultiLine"
    android:lines="4"
    android:minLines="3"/>

<Button
    android:id="@+id/sendButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Send SMS"
```

```
        android:layout_marginTop="16dp"/>


</LinearLayout>
```

AndroidManifest.xml

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.notifications">


    <!-- Permissions -->
    <uses-permission android:name="android.permission.SEND_SMS"/>
    <uses-permission android:name="android.permission.RECEIVE_SMS"/>
    <uses-permission android:name="android.permission.READ_SMS"/>


    <application
        android:allowBackup="true"
        android:label="Notifications"
        android:icon="@mipmap/ic_launcher"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
```

```
    android:theme="@style/Theme.Notifications">



<activity android:name=".MainActivity"
    android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.MAIN" />
        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
</activity>

<!-- BroadcastReceiver for receiving SMS -->
<receiver android:name=".SmsReceiver" android:exported="true">
    <intent-filter>
        <action android:name="android.provider.Telephony.SMS_RECEIVED" />
    </intent-filter>
</receiver>
```

```
</application>
    </manifest>
```

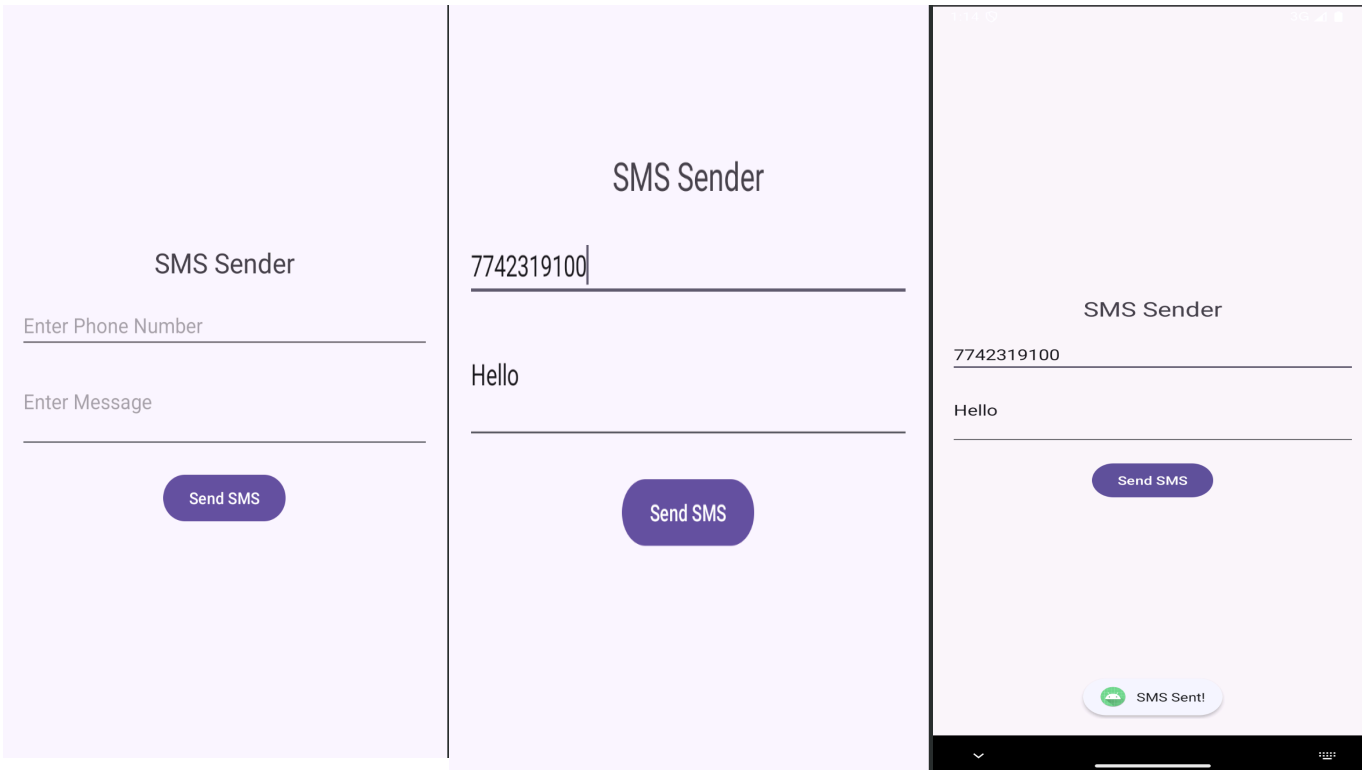
SmsReceiver.java

```
package com.example.notifications;


import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.os.Bundle;
import android.telephony.SmsMessage;
import android.widget.Toast;


public class SmsReceiver extends BroadcastReceiver {
    @Override
    public void onReceive(Context context, Intent intent) {
        Bundle bundle = intent.getExtras();
        if (bundle != null) {
            Object[] pdus = (Object[]) bundle.get("pdus");
```

```
            if (pdus != null) {
                for (Object pdu : pdus) {
                    SmsMessage smsMessage = SmsMessage.createFromPdu((byte[]) pdu);
                    String sender = smsMessage.getDisplayOriginatingAddress();
                    String message = smsMessage.getDisplayMessageBody();
                    Toast.makeText(context, "SMS From: " + sender + "\n" + message,
Toast.LENGTH_LONG).show();
                }
            }
        }
    }
}
```



Question 5

MainActivity.java

```
package com.example.imagecap;


import android.content.Intent;

import android.graphics.Bitmap;

import android.os.Bundle;

import android.os.Environment;

import android.provider.MediaStore;

import android.view.View;

import android.widget.Button;

import android.widget.ImageView;

import android.widget.Toast;


import androidx.annotation.Nullable;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.app.ActivityCompat;

import androidx.core.content.ContextCompat;
```

```
import android.Manifest;
import android.content.pm.PackageManager;


import java.io.File;
import java.io.FileOutputStream;
import java.io.IOException;


public class MainActivity extends AppCompatActivity {


    private static final int CAMERA_REQUEST_CODE = 100;
    private static final int CAMERA_PERMISSION_CODE = 200;


    ImageView imageView;
    Button captureButton;


    @Override
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        imageView = findViewById(R.id.imageView);
        captureButton = findViewById(R.id.captureButton);

        captureButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                if (ContextCompat.checkSelfPermission(MainActivity.this,
Manifest.permission.CAMERA) != PackageManager.PERMISSION_GRANTED
                        || ContextCompat.checkSelfPermission(MainActivity.this,
Manifest.permission.WRITE_EXTERNAL_STORAGE) !=
PackageManager.PERMISSION_GRANTED) {
                    ActivityCompat.requestPermissions(MainActivity.this,
                            new String[]{Manifest.permission.CAMERA,
Manifest.permission.WRITE_EXTERNAL_STORAGE},
                            CAMERA_PERMISSION_CODE);
                } else {
```

```
                openCamera();
            }
        }
    });
}

private void openCamera() {
    Intent intent = new Intent(MediaStore.ACTION_IMAGE_CAPTURE);
    startActivityForResult(intent, CAMERA_REQUEST_CODE);
}

@Override
protected void onActivityResult(int requestCode, int resultCode, @Nullable Intent data) {
    super.onActivityResult(requestCode, resultCode, data);

    if (requestCode == CAMERA_REQUEST_CODE && resultCode == RESULT_OK &&
data != null) {
        Bitmap photo = (Bitmap) data.getExtras().get("data");
        imageView.setImageBitmap(photo);
```

```
            saveImageToStorage(photo);
        }
    }

    private void saveImageToStorage(Bitmap bitmap) {
        File path =
Environment.getExternalStoragePublicDirectory(Environment.DIRECTORY_PICTURES);
        File file = new File(path, "captured_image_" + System.currentTimeMillis() + ".jpg");

        try {
            path.mkdirs();
            FileOutputStream out = new FileOutputStream(file);
            bitmap.compress(Bitmap.CompressFormat.JPEG, 100, out);
            out.close();
            Toast.makeText(this, "Image saved to Pictures", Toast.LENGTH_SHORT).show();
        } catch (IOException e) {
            e.printStackTrace();
            Toast.makeText(this, "Failed to save image", Toast.LENGTH_SHORT).show();
        }
```

```
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="300dp"
        android:layout_height="300dp"
        android:scaleType="centerCrop"
        android:layout_marginBottom="16dp"
        android:src="@mipmap/ic_launcher" />
```

```
    <Button
        android:id="@+id/captureButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Capture Image" />
</LinearLayout>
```

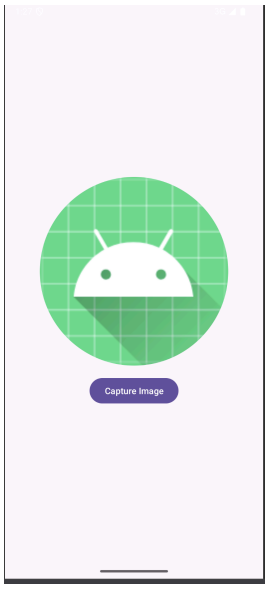
AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.imagecap">

    <!-- Permissions -->
    <uses-permission android:name="android.permission.CAMERA"/>
    <uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
    <uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE"/>

    <application
```

```
        android:allowBackup="true"
        android:label="ImageCap"
        android:icon="@mipmap/ic_launcher"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.ImageCap">
        <activity android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>
                <category android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>
    </application>
</manifest>
```



Question 6

MainActivity.java

```
package com.example.emailapp;


import android.content.Intent;

import android.os.Bundle;

import android.widget.Button;

import android.widget.EditText;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {

    EditText Email, Subject, Message;

    Button button;


    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);
```

```
setContentView(R.layout.activity_main);


Email = findViewById(R.id.Email);

Subject = findViewById(R.id.Subject);

Message = findViewById(R.id.Message);

button = findViewById(R.id.button);


button.setOnClickListener(v -> {

    String email = Email.getText().toString();

    String subject = Subject.getText().toString();

    String message = Message.getText().toString();


    Intent intent = new Intent(Intent.ACTION_SEND);

    intent.setType("message/rfc822");

    intent.putExtra(Intent.EXTRA_EMAIL, new String[]{email});

    intent.putExtra(Intent.EXTRA_SUBJECT, subject);

    intent.putExtra(Intent.EXTRA_TEXT, message);
```

```
            try {
                startActivity(Intent.createChooser(intent, "Choose Email App"));
            } catch (android.content.ActivityNotFoundException ex) {
                Toast.makeText(MainActivity.this, "No email client installed.",
Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#F5F5F5"
    android:orientation="vertical"
```

```
android:padding="16dp"

tools:context=".MainActivity">


<TextView

    android:id="@+id/appName"

    android:layout_width="match_parent"

    android:layout_height="wrap_content"

    android:layout_gravity="center"

    android:gravity="center_horizontal"

    android:padding="10dp"

    android:text="Email App"

    android:textColor="@android:color/black"

    android:textSize="25sp" />


<EditText

    android:id="@+id/Email"

    android:layout_width="match_parent"

    android:layout_height="wrap_content"
```

```
    android:hint="Recipient Email"
    android:inputType="textEmailAddress"
    android:textColor="#343A40"
    android:textColorHint="#6C757D" />

<EditText
    android:id="@+id/Subject"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:hint="Subject"
    android:textColor="#343A40"
    android:textColorHint="#6C757D" />

<EditText
    android:id="@+id/Message"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
```

```
        android:layout_marginTop="10dp"
        android:gravity="top"
        android:hint="Message"
        android:inputType="textMultiLine"
        android:minLines="5"
        android:textColor="#343A40"
        android:textColorHint="#6C757D" />

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center_horizontal"
        android:layout_marginTop="20dp"
        android:text="Send Email" />

</LinearLayout>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    package="com.example.emailapp">

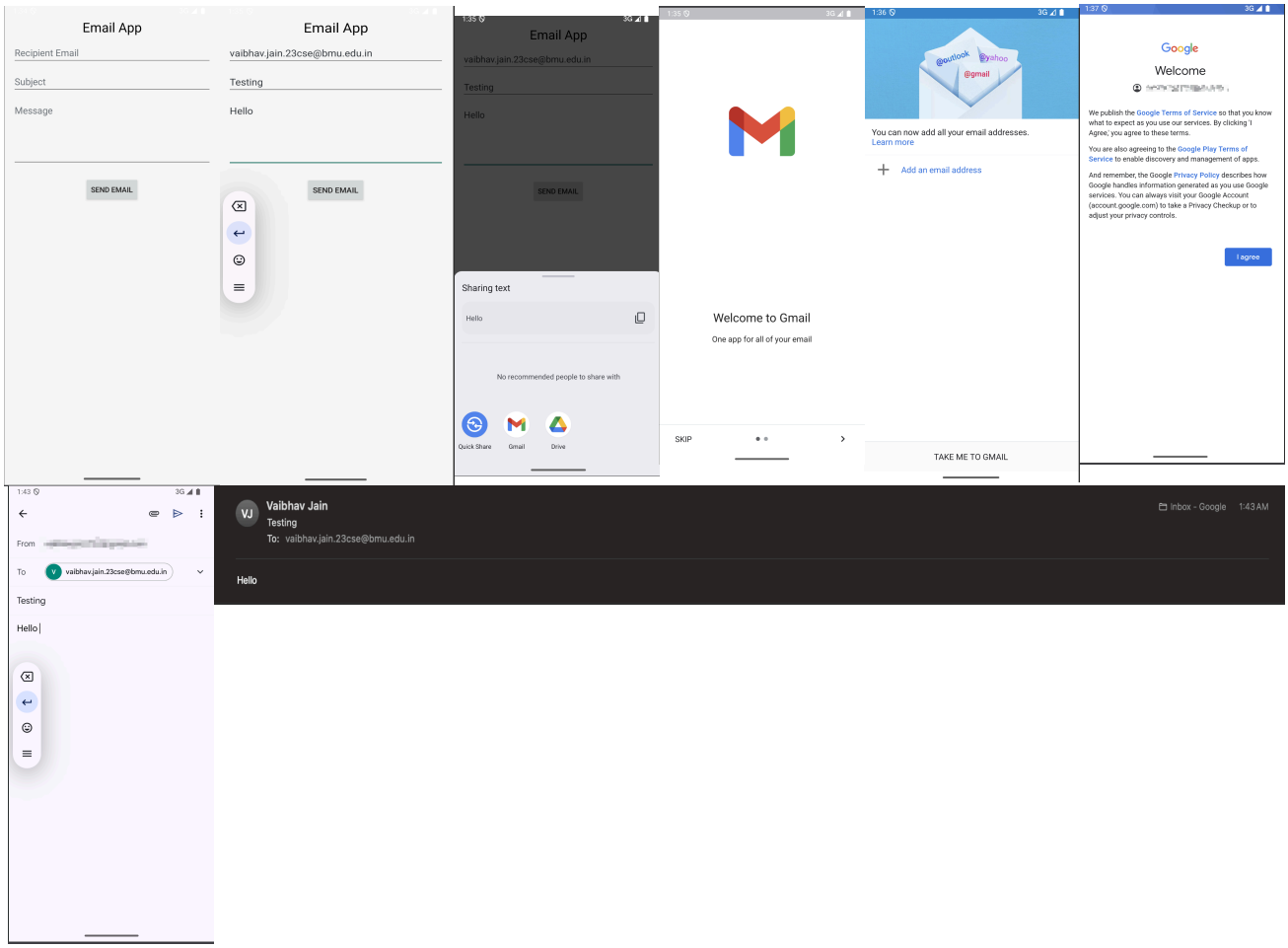
    <application
        android:allowBackup="true"
        android:label="EmailApp"
        android:icon="@mipmap/ic_launcher"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppCompat.Light.NoActionBar"
        tools:targetApi="31">

        <activity
            android:name=".MainActivity"
            android:exported="true">
```

```
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>

</application>

</manifest>
```



Question 7

MainActivity.java

```
package com.example.callmate;


import android.Manifest;

import android.content.Intent;

import android.content.pm.PackageManager;

import android.net.Uri;

import android.os.Bundle;

import android.view.View;

import android.widget.Button;

import android.widget.EditText;

import android.widget.Toast;


import androidx.annotation.NonNull;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.app.ActivityCompat;

import androidx.core.content.ContextCompat;
```

```
public class MainActivity extends AppCompatActivity {
    private static final int REQUEST_CALL_PERMISSION = 1;
    private EditText phoneNumberEditText;
    private Button callButton;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        phoneNumberEditText = findViewById(R.id.phoneNumberEditText);
        callButton = findViewById(R.id.callButton);

        callButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                makePhoneCall();
```

```
        }
    });
}

private void makePhoneCall() {
    String phoneNumber = phoneNumberEditText.getText().toString();

    if (phoneNumber.trim().length() > 0) {
        if (ContextCompat.checkSelfPermission(MainActivity.this,
                Manifest.permission.CALL_PHONE) !=
PackageManager.PERMISSION_GRANTED) {
            ActivityCompat.requestPermissions(MainActivity.this,
                new String[]{Manifest.permission.CALL_PHONE},
                REQUEST_CALL_PERMISSION);
        } else {
            String dial = "tel:" + phoneNumber;
            startActivity(new Intent(Intent.ACTION_CALL, Uri.parse(dial)));
        }
    } else {
```

```
            Toast.makeText(MainActivity.this, "Please enter a phone number",
Toast.LENGTH_SHORT).show();
        }
    }


    @Override
    public void onRequestPermissionsResult(int requestCode, @NonNull String[] permissions,
                                           @NonNull int[] grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults);
        if (requestCode == REQUEST_CALL_PERMISSION) {
            if (grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
                makePhoneCall();
            } else {
                Toast.makeText(this, "Permission DENIED", Toast.LENGTH_SHORT).show();
            }
        }
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#F2EFE4"
    android:gravity="center_horizontal"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="CallMate"
        android:textSize="24sp"
```

```
    android:textStyle="bold"
    android:textColor="#000000"
    android:layout_marginBottom="24dp"/>

<ImageView
    android:id="@+id/imageView"
    android:layout_width="200dp"
    android:layout_height="200dp"
    android:layout_marginBottom="24dp"
    app:srcCompat="@drawable/ic_launcher_foreground" />

<EditText
    android:id="@+id/phoneNumberEditText"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Phone Number"
    android:inputType="phone"
    android:padding="12dp"
```

```
        android:textColor="#000000"
        android:textColorHint="#666666"
        android:layout_marginBottom="16dp" />

    <Button
        android:id="@+id/callButton"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Make Call"
        android:backgroundTint="#4CAF50"
        android:textColor="#FFFFFF"
        android:padding="12dp" />
</LinearLayout>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.callmate"
    xmlns:tools="http://schemas.android.com/tools">
```

```
<uses-feature android:name="android.hardware.telephony" android:required="false" />
<uses-permission android:name="android.permission.CALL_PHONE" />
<uses-permission android:name="android.permission.READ_PHONE_STATE" />

<application
    android:allowBackup="true"
    android:icon="@drawable/ic_launcher_foreground"
    android:label="CallMate"
    android:roundIcon="@drawable/ic_launcher_foreground"
    android:supportsRtl="true"
    android:theme="@style/Theme.CallMate"
    tools:targetApi="31">

    <activity android:name=".MainActivity" android:exported="true">
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
```

```
            </intent-filter>
        </activity>

        <receiver android:name=".PhoneCallReceiver" android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.PHONE_STATE" />
            </intent-filter>
        </receiver>

    </application>
</manifest>
```

PhoneCallReceiver.java

```
package com.example.callmate;

import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.telephony.TelephonyManager;
```

```
import android.widget.Toast;


public class PhoneCallReceiver extends BroadcastReceiver {
    @Override
    public void onReceive(Context context, Intent intent) {
        if (intent.getAction().equals(TelephonyManager.ACTION_PHONE_STATE_CHANGED))
{
            String phoneState = intent.getStringExtra(TelephonyManager.EXTRA_STATE);
            String phoneNumber =
intent.getStringExtra(TelephonyManager.EXTRA_INCOMING_NUMBER);

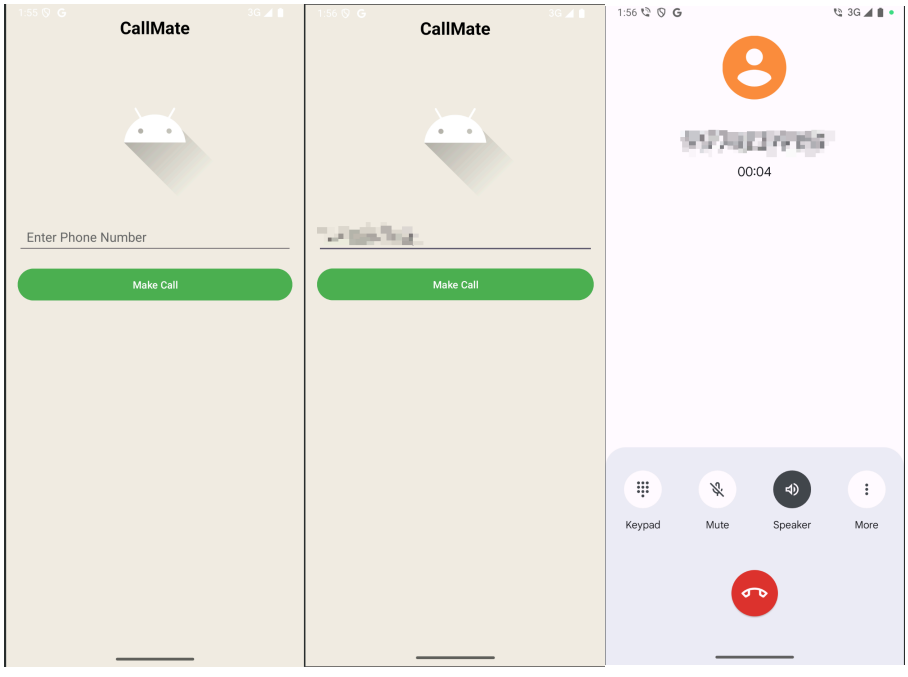

            if (phoneState.equals(TelephonyManager.EXTRA_STATE_RINGING)) {
                showToast(context, "Incoming Call From: " + (phoneNumber != null ? phoneNumber :
"Unknown"));
            } else if (phoneState.equals(TelephonyManager.EXTRA_STATE_OFFHOOK)) {
                showToast(context, "Call Started");
            } else if (phoneState.equals(TelephonyManager.EXTRA_STATE_IDLE)) {
                showToast(context, "Call Ended");
            }
```

```
        }
    }

    private void showToast(Context context, String message) {
        Toast.makeText(context, message, Toast.LENGTH_LONG).show();
    }
}
```

strings.xml

```
<resources>
    <string name="app_name">CallMate</string>
    <string name="calling_applicaton">CallMate</string>
    <string name="enter_phone_number">Enter Phone Number</string>
    <string name="make_call">Make Call</string>
</resources>
```



Question 8

MainActivity.java

```
package com.example.smsmessenger;


import android.Manifest;

import android.content.pm.PackageManager;

import android.os.Bundle;

import android.telephony.SmsManager;

import android.view.View;

import android.widget.Button;

import android.widget.EditText;

import android.widget.Toast;


import androidx.annotation.NonNull;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.app.ActivityCompat;

import androidx.core.content.ContextCompat;
```

```
public class MainActivity extends AppCompatActivity {

    private static final int SMS_PERMISSION_REQUEST_CODE = 1;
    private EditText phoneNumberEditText, messageEditText;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        phoneNumberEditText = findViewById(R.id.phoneNumberEditText);
        messageEditText = findViewById(R.id.messageEditText);
        Button sendSMSButton = findViewById(R.id.sendSMSButton);

        sendSMSButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                sendSMS();
```

```
        }
    });
}

private void sendSMS() {
    String phoneNumber = phoneNumberEditText.getText().toString();
    String message = messageEditText.getText().toString();

    if (ContextCompat.checkSelfPermission(this, Manifest.permission.SEND_SMS) !=
PackageManager.PERMISSION_GRANTED) {
        ActivityCompat.requestPermissions(this, new
String[]{Manifest.permission.SEND_SMS}, SMS_PERMISSION_REQUEST_CODE);
    } else {
        SmsManager smsManager = SmsManager.getDefault();
        smsManager.sendTextMessage(phoneNumber, null, message, null, null);
        Toast.makeText(this, "SMS sent!", Toast.LENGTH_SHORT).show();
    }
}
```

```
    @Override
    public void onRequestPermissionsResult(int requestCode, @NonNull String[] permissions,
@NonNull int[] grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults);
        if (requestCode == SMS_PERMISSION_REQUEST_CODE) {
            if (grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
                sendSMS();
            } else {
                Toast.makeText(this, "Permission denied. Cannot send SMS.",
Toast.LENGTH_SHORT).show();
            }
        }
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
```

```
android:layout_height="match_parent"

android:orientation="vertical"

android:padding="16dp">


<EditText

    android:id="@+id/phoneNumberEditText"

    android:layout_width="match_parent"

    android:layout_height="wrap_content"

    android:hint="Enter phone number"

    android:inputType="phone" />


<EditText

    android:id="@+id/messageEditText"

    android:layout_width="match_parent"

    android:layout_height="wrap_content"

    android:hint="Enter message"

    android:inputType="text" />
```

```
    <Button
        android:id="@+id/sendSMSButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send SMS" />

</LinearLayout>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    package="com.example.smsmessenger">

    <application
        android:allowBackup="true"
        android:label="@string/app_name"
        android:supportsRtl="true"
        android:theme="@style/Theme.SMSMessenger"
```

```
tools:targetApi="31">


<activity
    android:name=".MainActivity"
    android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.MAIN" />
        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
</activity>


<receiver
    android:name=".SMSReceiver"
    android:exported="false">
    <intent-filter>
        <action android:name="android.provider.Telephony.SMS_RECEIVED" />
    </intent-filter>
</receiver>
```

```
    </application>


</manifest>
```

SMSReceiver.java

```
package com.example.smsmessenger;


import android.content.BroadcastReceiver;

import android.content.Context;

import android.content.Intent;

import android.telephony.SmsMessage;


public class SMSReceiver extends BroadcastReceiver {


    @Override

    public void onReceive(Context context, Intent intent) {

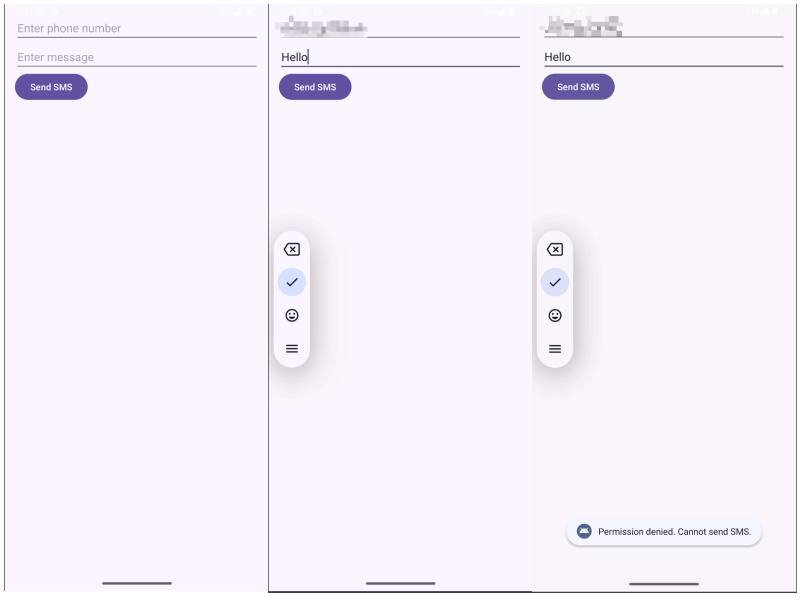
        if (intent.getAction().equals("android.provider.Telephony.SMS_RECEIVED")) {

            Object[] pdus = (Object[]) intent.getExtras().get("pdus");
```

```
            StringBuilder messageBuilder = new StringBuilder();
            for (Object pdu : pdus) {
                SmsMessage smsMessage = SmsMessage.createFromPdu((byte[]) pdu);
                String message = smsMessage.getMessageBody();
                messageBuilder.append(message).append("\n");
            }
            // Show received message to the user (in your activity or use a BroadcastReceiver for the
UI)
        }
    }
}
```

strings.xml

```
<resources>
    <string name="app_name">SMSMessenger</string>
    <string name="enter_phone_number">Enter Phone Number</string>
    <string name="enter_message">Enter Message</string>
    <string name="send_sms">Send SMS</string>
    <string name="received_sms">Received SMS</string>
</resources>
```


Enter phone number
Enter message
Send SMS
Hello
Send SMS
Hello
Send SMS
Permission denied. Cannot send SMS.